



Influence of environmental conditions on the ecological quality of wood biomass growing on different soils and geological background

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Abstract

Purpose The aim of this study was to evaluate the influence of environmental factors, particularly the geological substrate, soil properties, and atmospheric pollution, on the elemental composition of ash produced from wood biomass combustion. The study also assessed the implications of these factors for the use of wood biomass as a high-quality energy resource and for the safe utilization or disposal of the resulting ash.

Materials and methods The study was conducted in six forest areas in Poland, differing in soil types (natural and technogenic), geological substrates (sands, ultrabasic rocks, post-industrial waste), and historical dust deposition rates. Two coniferous and two deciduous tree species were selected from each area, from which samples of wood, bark, and small twigs were collected. Soil samples were collected as cores to a depth of 30 cm. Biomass samples were dried, crushed, and combusted in stages at controlled temperatures. The elemental composition of the ash was analyzed using ICP-OES and ICP-MS, determining macro- and microelements as well as potentially toxic elements. The results were verified against reference materials and repeated three times. The elemental composition of the soil was determined using XRF, and its pH was measured potentiometrically.

Results and discussion The chemical composition of wood biomass ash varies depending on species, plant part, and location. In most cases, Ca was the dominant element, with K and Mg occurring in variable proportions. Bark and twigs often contained more Fe, Al, and trace elements than wood. High concentrations of Zn were noted particularly in birch, while Cd and Pb were elevated in many samples, especially in contaminated areas and on technogenic soils, with Cd accumulating more readily in wood than Pb.

Conclusions Environmental conditions and tree species determine the quality of biomass as fuel and the potential use of ash for soil fertilization. Wood reflects soil composition, while bark, twigs, and needles accumulate air pollutants, particularly in coniferous species. Anthropogenic elements (Pb, Cd, Sn, Sb) accumulate primarily in above-ground parts, but can also originate from the soil. Cd is more mobile than lead and accumulates more readily. Birch hyperaccumulates Zn, which makes its ash unsuitable for use as a fertilizer. Most ash samples did not meet fertilization standards.

Keywords Wood biomass · Wood ash · Soil pollution · Air pollution · Environmental assessment

1 Introduction

Biomass is widely considered a potential renewable energy source and an alternative to conventional fossil fuels. Woody biomass, in particular, is available in a variety of forms and is often regarded as CO₂-neutral in energy policies and corporate communications. However, the assumption that wood combustion is inherently carbon-neutral has been increasingly questioned by scientists and environmental organizations. The environmental impacts of woody biomass combustion depend largely on the quality and origin of the biomass (Saravanakumar et al. 2023). The ecological

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quality of woody biomass, like food, depends on the conditions in which it is grown. Biomass production often uses degraded lands unsuitable for producing safe food (Evangelou et al. 2012; Sheikh and Abhilash 2016). Could non-food agriculture, which often uses contaminated soils, be a promising area for energy crops? It is obvious that the amount of available nutrients (e.g. nitrogen, phosphorus, potassium, calcium, magnesium) in the soil directly affects the chemical composition of wood, leaves, twigs and bark (Osman 2013; Bhatla and Lal 2023). Their uptake is controlled by soil pH. Soil acidification may limit the availability of certain elements (e.g. calcium, magnesium) and increase the mobility of heavy metals (e.g. cadmium, lead, aluminum), which may accumulate in biomass (Yadav et al. 2020). Also soil type and structure play the important role. Clay and loam soils may have a higher sorption capacity, which affects the ability to retain and make nutrients available (Kome et al. 2019).

Many tree species such as: Scots pine (*Pinus sp.*), European larch (*Larix sp.*), European oak (*Quercus robur L.*), alder (*Alnus sp.*) poplar (*Populus L*) and lime tree (*Tilia sp.*) are used in the reclamation of mining and processing waste dumps (Ranjan et al. 2016; Pietrzykowski 2019; Vachova et al. 2021). Often, they are used solely to create green spaces with a park-like or forest character, but in some post-industrial regions there have also been used for establishing energy crops in these areas. Many tree species characterized by rapid growth, large biomass, and high resistance to environmental stress seem to be ideal raw materials for producing woody biomass for energy purposes (Bungart and Hüttl 2001; Menéndez and Loredó 2018; Tran et al. 2022). Also, areas with current and historical air pollution, especially related to the deposition of industrial and urban dust, may have a negative impact on the ecological quality of biomass used for energy purposes, even though it grows on natural soils (Evangelou et al. 2012).

Under certain specific geological conditions, a natural geological background rich in metals may also influence the elevated content of trace elements in wood biomass (Reimann et al. 2007a; 2007b). The importance of geological background was confirmed during the study in Scandinavia (Reimann et al., 2001). The authors reported a strong effect of Cambro-Silurian sandstone and shales on the concentrations of Ca, Ba, and Ni in birch leaves, bark, and wood (Reimann et al. 2001, 2007a). They concluded that, with the exception of these three elements, each investigated tree species (birch, mountain ash, and spruce) exhibited unique uptake characteristics, reacted differently, and displayed species-specific uptake (or exclusion) patterns (Reimann et al. 2007b). However, the effect of species on the elemental composition of wood ash from trees growing under different environmental conditions is not well understood.

The fundamental question is whether the combustion of biomass obtained from such areas is safe for the environment, and whether the solid combustion products can be safely reused in the economy and the environment. The primary objective of this study was to determine how environmental factors, including the geological substrate, soil characteristics, and atmospheric pollution, affect the elemental composition of ash produced during wood biomass combustion. Understanding these relationships is essential for evaluating the suitability of wood biomass as a high-quality energy source and for ensuring the safe management and potential utilization of the resulting ash.

2 Materials and methods

2.1 Study area

The study areas comprised six forested sites located in different parts of Poland (Fig. 1). An important criterion for site selection was that trees grew on diverse natural and anthropogenic substrates, including Podzols developed on Quaternary sands, Cambisols developed on alkaline metamorphic rocks (serpentinite), Rendzinas developed on ore-bearing dolomites, and initial soils formed on anthropogenic substrates in metal ore mining and/or processing dumps. These areas were also characterized by varying levels of historical and current soil and air pollution, as well as by different sources emitting a wide range of elements (Tables 1 and 2). A forest area located in northern Poland, within the Przasnysz Forest District, was selected as the reference site. It is situated far from local emission sources, in a region characterized by relatively low atmospheric deposition and good air quality (Airly 2026). The selected tree stand grows on a Podzol developed on Quaternary sands with low concentrations of potentially toxic elements (PTEs) in the organic and eluvial soil horizons (Table 3).

2.2 Material and sampling procedure

At each site, two coniferous and two deciduous tree species were selected from the dominant tree stand. Silver birch (*Betula pendula* Roth) and Scots pine (*Pinus sylvestris L.*) were present at all sites, while European larch (*Larix decidua* Mill.) and pedunculate oak (*Quercus robur L.*) were found at most sites. At the Złoty Stok site, larch and oak were not present in the tree stand; therefore, Norway spruce (*Picea abies L.*) and European aspen (*Populus tremula L.*) were selected instead. At the Nowy Dwór site, white poplar (*Populus alba L.*) was selected instead of oak, as oak was not present there. Wood samples were collected from one freshly cut tree of each selected species. Additionally, bark

Fig. 1 Location of sampling sites described in Table 1: 1- Nowy Dwór; 2- Łosień; 3-Ząbkowice; 4-Radunia; 5-Złoty Stok; 6 - Przasnysz



and twigs were sampled separately. In the case of coniferous species, the twig samples also included needles. The collected material was transported to the laboratory, air-dried at room temperature, and chipped using ceramic knives. Three independent subsamples of each material were then prepared for combustion. To assess the level of soil pollution, eight cores (0–30 cm) were collected at each wood biomass sampling site in accordance with the ISO 21226:2019 procedure (International Organization for Standardization 2019).

2.3 Combustion of biomass samples and chemical analyses

Crushed biomass samples were combusted under controlled laboratory conditions in a muffle furnace. To avoid rapid ignition and premature sintering of elements with lower melting points (e.g. K and Na), and to ensure complete combustion of organic compounds, the process was carried out in four stages at temperatures of 200 ± 15 °C, 300 ± 15 °C, 450 ± 15 °C, and 850 ± 15 °C.

The contents of 14 macro- and microelements (Al, Fe, P, S, Mg, Na, K, Ca, Ti, Ba, Mn, Cu, Pb, and Zn) in the ash were determined by inductively coupled plasma optical emission spectrometry (ICP-OES - Avio 200, Perkin Elmer) after digestion in aqua regia. The same digestion procedure was applied for the determination of potentially toxic elements (PTEs), which were quantified using the more sensitive inductively coupled plasma mass spectrometry (ICP-MS) technique. Measurements were performed using an Elan DRC-e mass spectrometer (PerkinElmer, USA), with instrument performance verified using ICP Quality Control Standard 1 (AccuStandard, New Haven, CT, USA). The following PTEs were analyzed: V, Co, Cd, Ni, W, Au, Mo, Sb, Sn, and Cr.

The accuracy of the analytical results was verified using certified reference materials. All measurements were performed in triplicate, and average values were used in the study. Because aqua regia digestion does not ensure complete decomposition of silicates, Si was not included among the analyzed macroelements. Volatile elements such as oxygen, hydrogen, nitrogen, and chlorine, which are released

Table 1 Description of investigated sites

Name of the site	Location	Site description	Tree species sampled	Level of pollution
Nowy Dwór	Bytom, Silesian Voivodeship, Poland	old dump of lead and zinc ore mining wastes	birch, larch, poplar, pine	very high
Łosień	Dąbrowa Górnicza (Łosień), Silesian Voivodeship, Poland	close to large metallurgical and coke plant, natural (Podzolic) sandy soil	birch, larch, oak, pine	very high
Ząbkowice	Dąbrowa Górnicza (Ząbkowice), Silesian Voivodeship, Poland	close to the dolomite quarry, natural Rendzina developed on ore-bearing dolomite	birch, larch, oak, pine	high
Radunia	Radunia Nature Reserve, Ślęża Massif, Lower Silesian Voivodeship, Poland	natural District Cambisol developed on serpentine rich in magnetite	birch, larch, oak, pine	medium
Złoty Stok	Złoty Stok, Lower Silesian Voivodeship, Poland	old dump of arsenic mining and processing wastes	birch, spruce, aspen, pine	medium
Przasnysz	Przasnysz, Masovian Voivodeship, north-central Poland	natural sandy soil, (Podzolic) (reference area)	birch, larch, oak, pine	low

as gases during combustion (high temperature) and sample preparation (drying, calcination), were not included in the analyses.

The content of PTEs in the soil substrate was estimated by analyzing collected soil and waste samples using X-ray fluorescence (XRF) with a portable Explorer 7000 spectrometer (pXRF), following the laboratory measurement procedure described by Magiera and Szuszkiewicz (2025). The soil reaction (pH in H₂O) was measured potentiometrically by the laboratory pH-metre in a suspension (1:5 mass to- volume ratio).

3 Results

3.1 Soil analyses

The examined soils exhibited a wide range of pH values, from alkaline Rendzina developed on dolomite in Ząbkowice (pH

Table 2 Description of investigated sites

Area	Historical periods	Estimated dust deposition (g/m ² ·year)	Main sources of pollution
Nowy Dwór ¹⁾	1970–1980 1990–2000 2010–2020	300–1000+ 50–200 <50	Zn–Pb heaps, local industrial and urban emissions
Łosień ¹⁾	1970–1980 1990–2000 2010–2020	300–1000+ 50–200 <50	steelworks and coking plant, continuous dust emissions since 1975
Ząbkowice ¹⁾	1970–1980 1990–2000 2010–2020	200–500 50–200 <50	moderate deposition from industrial source (steelwork) and dusting from local dolomite quarry
Radunia ²⁾	1970–1980 1990–2000 2010–2020	20–100 10–40 2–20	long range transport (lack of local sources)
Złoty Stok ²⁾	XIX–XX century/up to 1960s 1990–2000 2010–2020	80–200 60–100 40–70	arsenic ore mining and processing from the end of the 19th century to the 1960s
Przasnysz ³⁾	current	<30	small local emission, Masovia background

¹⁾ Regional environmental monitoring reports of the former State Inspectorate for Environmental Protection (PIOS) and later WIOŚ Katowice (1980s–2020)

²⁾ Regional environmental monitoring reports (WIOŚ Wrocław/GIOS) for Lower Silesia

³⁾ Regional environmental monitoring reports (WIOŚ Warszawa/GIOS) for Mazovia

up to 8.0), through slightly alkaline technogenic soils developed on mining and processing waste and the slightly alkaline soils of Łosień (pH 7.5), where prolonged atmospheric deposition of industrial dust has occurred, to acidic soils in Przasnysz and strongly acidic soils in the Radunia Reserve (pH 4.8). The highest PTE concentrations, reaching up to 10,000 mg/kg of Zn and Pb, were recorded in the waste from the Nowy Dwór dumping ground. Significantly lower, but still potentially hazardous, concentrations of these two metals (>1,000 mg/kg) were found in natural soils developed on ore-bearing dolomites in Ząbkowice and, in the case of Zn, also in the heavily contaminated natural soil at Łosień. Pb concentrations exceeding 900 mg/kg were also recorded in the waste from Złoty Stok, where they were accompanied by elevated Cr (870 mg/kg) and Ni (430 mg/kg) concentrations. High concentrations of these metals (>250 mg/

Table 3 Content of some elements in topsoil cores (0–30 cm) measured by portable XRF

Sampling place and soil horizons included		pH	Fe ₂ O ₃ %	Mn mg/kg	Cr	Ni	Zn	Pb	Ti	V
Nowy Dwór (<i>n</i> =8) Technosol (A/C and C)	min.	7.2	2.51	5007	32	23	8250	7560	53	53
	max.	8.0	5.89	5873	82	45	10,512	11,875	103	121
	mean	7.6	4.20	5455	57	34	9560	7139	78	87
Łosień (<i>n</i> =8) Podzol (Ofh and Ees)	min.	7.0	2.47	918	38	10	745	119	3083	63
	max.	7.9	3.60	1150	72	37	1103	193	4042	96
	mean	7.5	3.27	1079	53	25	999	174	3476	82
Ząbkowice (<i>n</i> =8) Rendzina (A and Abr)	min.	7.8	5.59	1339	84	35	1542	1176	3668	88
	max.	8.1	6.88	1578	94	59	3220	2663	5189	117
	mean	8.0	6.20	1461	89	46	2177	1707	4297	103
Radunia (<i>n</i> =8) District Cambisol (A)	min.	4.0	3.70	394	208	246	81	234	2302	14
	max.	5.6	3.88	446	286	369	109	273	2592	64
	mean	4.8	3.77	415	266	300	103	261	2439	39
Złoty Stok (<i>n</i> =8) Technosol (A/C and C)	min.	7.4	8.29	3060	862	326	515	821	37,662	1089
	max.	7.6	9.42	8333	887	486	622	1061	38,567	1143
	mean	7.5	8.83	5459	870	431	586	947	38,136	1112
Przasnysz (<i>n</i> =8) Podzol (Ofh A and Ees)	min.	4.8	0.09	20	<18	10.9	<7.5	<3.5	557	13
	max.	5.6	0.41	208	<18	21.4	<7.5	<3.5	2303	78
	mean	5.2	0.15	118	<18	18.0	<7.5	<3.5	1200	36

kg) were also observed in the natural soil of the Radunia Reserve, developed on serpentinite naturally enriched in Cr and Ni. The strongly acidic soil reaction (pH<5.0) may promote the mobilization of these metals. The highest Mn concentrations (>5,000 mg/kg) were recorded in both types of waste underlying the technogenic soils at Nowy Dwór and Złoty Stok. The lack of significant emitters of dust pollution results in low metal values in the uppermost horizons (0–30 cm). In this area the analyzed metals occurred in the upper soil horizons at low values (Table 3).

3.2 Nowy Dwór

In birch, Ca was the dominant macroelement in wood (>400 g/kg; 72%), while K was <12% (>66 g/kg). K content increased in bark (265 g/kg; 46%) and twigs (>335 g/kg; 74%) and was the highest among all samples. Mg was relatively stable (29–30 g/kg in bark and twigs; 5.3–6.2%) and reached up to 35 g/kg in wood. Zn was unusually high in wood (>40 g/kg; 7%) but much lower in bark and twigs (4.5–7 g/kg). Pb was 1470 mg/kg in wood and lower in bark and twigs (230 and 101 mg/kg) (Fig. 2, first row, Table S1). Cd was <60 mg/kg, and Mo was ~7 mg/kg (Table 4).

In larch, Ca was more evenly distributed: 385 g/kg in wood (69%) and 265–282 g/kg in twigs and bark (54–57%). In wood, K and Mg were similar (~55 g/kg each), and together with Ca accounted for ~90% of the inorganic fraction. In bark and twigs, Fe was high (100 and 92 g/kg; ~20%), while Al reached 31–32 g/kg (~6.5%), with Mg, K, and S contributing ~15%. Zn was 12 g/kg in bark and 9 g/kg in twigs. Pb reached 5.7 g/kg in twigs, while wood

contained ~430 mg/kg (Fig. 2, second row, Table S1). Elevated trace elements included Sb and Sn, with the highest values recorded in bark (48 and 23.5 mg/kg) and twigs (68 and 32 mg/kg), respectively (Table 4).

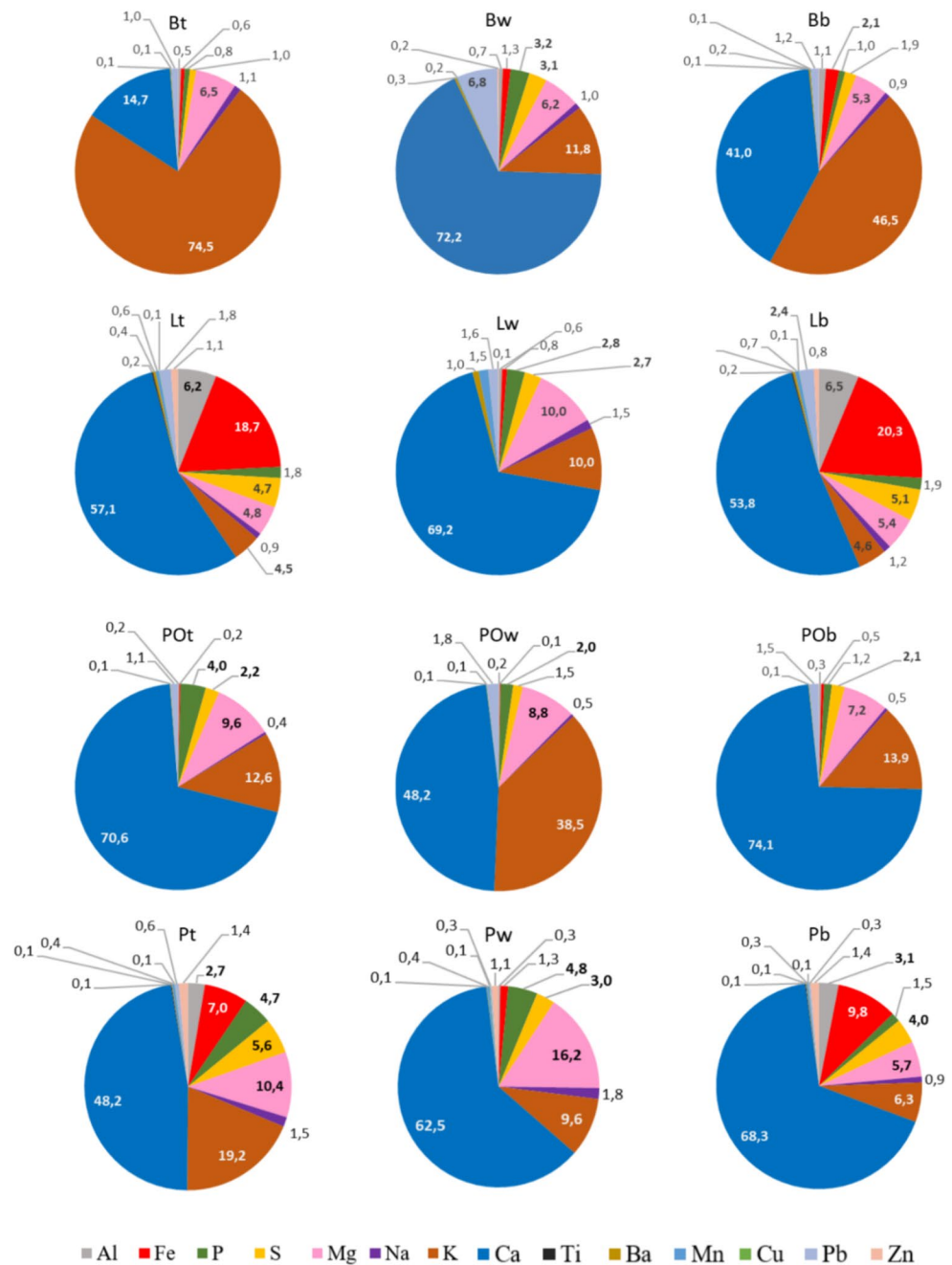
In poplar ash, Ca/K patterns differed from birch. In wood, Ca (256 g/kg; 48%) and K (204 g/kg; 38%) were comparable, while in bark and twigs Ca dominated (>70%; ~430 g/kg). K accounted for 12–14% and Mg for 7–9.5% across fractions. P ranged from 2 to 4% (10–24.5 g/kg), similar to birch wood (Fig. 2, third row, Table S1). Cd was elevated in wood and bark (>200 mg/kg) and ~175 mg/kg in twigs (Table 4).

In pine samples, Ca dominated ash from wood, bark, and twigs (48–68%; 250–320 g/kg). K content was highest in twigs (19%; ~99 g/kg), while in wood and bark it remained below 10%. Mg was relatively high in wood and twigs (80 and 53 g/kg; 16% and 10%). Fe was elevated in bark and twigs (7–10%; 35–45 g/kg), (Fig. 2, fourth row, Table S1) with similar Al contents (13–14 g/kg). Zn ranged from 5.5 to >7.5 g/kg. Pb reached 3.1 g/kg in twigs. Cu was high in wood (>550 mg/kg) (Table S1). Cd was very high (>290 mg/kg) but lower in bark (<130 mg/kg) and twigs (24 mg/kg). Elevated concentrations of other trace elements were also observed in pine bark ash, particularly Sb and Sn (26 and 19.5 mg/kg, respectively) (Table 4).

3.3 Łosień

In birch ash, Ca was the dominant macroelement, ranging from 425 g/kg in bark (77%) to 290 g/kg in wood (54%), with twigs at 358 g/kg (72%). K was the second most

Fig. 2 Chemical composition charts for ash samples after combustion of wood biomass collected in Nowy Dwór. Capital letters indicate tree species (B – birch, PO – poplar, L – larch, P – pine), whereas lowercase letters indicate the type of biomass (t – twigs, w – wood, b – bark). The percentage is related to the total mass of 14 investigated main components



abundant element (5.7–14.4%), while Mg in wood was similar to K (78.5 g/kg). P (>47 g/kg, ~9%) and S (~20 g/kg in wood) followed. Fe was significant mainly in bark (34 and 27.5 g/kg; >5–6%). Zn was consistently high across all fractions (~10 g/kg). Among trace elements, elevated Cu was noteworthy, primarily concentrated in wood (403 mg/kg) (Fig. 3, first row, Table S1)).

In the chemical composition of oak, Ca accounted for ~64% in wood and up to 85% in bark and twigs. Wood also contained K (14%), Mg (8.6%), S (4.7%), and P (3.9%). In bark and twigs, Fe was the second most abundant element (4–5.5%; 29 and 22 g/kg). Oak showed elevated Cu in wood

(425 mg/kg), with lower levels in other fractions (Fig. 3, second row, Table S1). Cr was highest in wood (158 mg/kg), followed by twigs (77 mg/kg) and bark (~34 mg/kg). Ni was elevated across all fractions, decreasing from ~47 mg/kg in wood to 32 mg/kg in twigs (Table 4).

In pine samples, Ca remained the dominant macronutrient, accounting for just over 50% in bark and 62.4% in twigs. Fe was the second most important element, reaching ~30% in bark, >17% in twigs, and 12% in wood (162, 95, and 74 g/kg, respectively). Pine wood also contained Mg (63 g/kg), K (51.5 g/kg), and S (36 g/kg) at levels >5%. In twig ash, S was the third most abundant element (51 g/

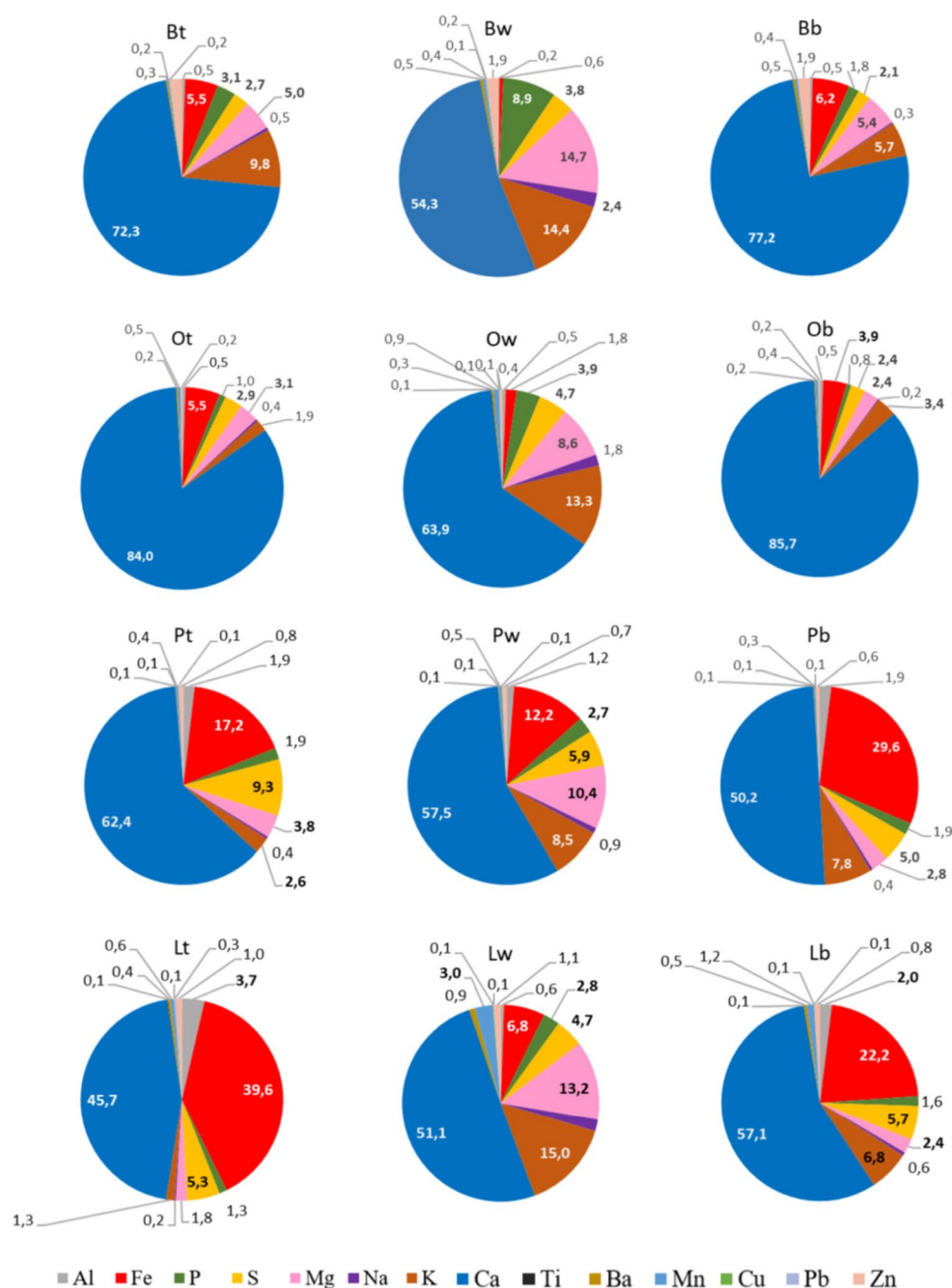
Table 4 Mean contents of trace element (in mg/kg) measured in ashes after combustion of biomass samples from investigated sites: t – in twigs, w – in wood, b- in bark

Sampling place	trees	V t; w; b	Co	Cd	Ni	W	Mo	Sb	Sn	Cr
Nowy Dwór (n=8)	birch	35.2; 38.5 ; 47.5	1.2; 4.0 ; 2.8	1.8; 57.8 ; 0.1	9.6; 53.4 ; 20.1	0.2; 0.8 ; 0.6	2.3; 7.0 ; 4.1	1.0; 9.6 ; 6.4	1.5; 8.6 ; 9.1	14.8; 36.9 ; 18.7
	poplar	18.4; 34.8 ; 33.1	1.3; 0.9 ; 1.6	176; 222 ; 220	17.2; 11.8 ; 17.5	0.1; 0.1 ; 0.2	1.3; 0.5 ; 0.5	2.5; 0.3 ; 3.8	1.8; 2.2 ; 4.2	6.2; 9.0 ; 10.4
	pine	7.6; <0.1 ; 36.2	2.1; 2.4 ; 8.7	24.8; 286 ; 127	12.3; 20.9 ; 49.6	0.9; 0.3 ; 2.8	2.8; 4.4 ; 8.4	9.2; 1.0 ; 26.0	8.9; 5.0 ; 19.5	17.0; 9.4 ; 68.0
	larch	70.6; 24.8 ; 59.4	12.1; 1.8 ; 12.7	80.8; 81.0 ; 122	70.6; 19.5 ; 69.7	3.8; 0.4 ; 3.5	10.5; 2.3 ; 9.2	68.1; 2.4 ; 48.0	31.9; 3.0 ; 23.5	98.2; 17.9 ; 97.3
	birch	12.2; <0.1 ; 14.4	3.6; 5.4 ; 5.8	25.2; 48.9 ; 16.9	27.6; 28.8 ; 28.0	12; 0.6 ; 1.2	1.9; 2.2 ; 1.3	1.9; 2.2 ; 1.3	3.1; 2.2 ; 1.7	49.9; 30.1 ; 39.3
Łosień (n=8)	oak	15.8; 4.9 ; 17.7	2.5; 2.5 ; 3.1	9.0; 14.4 ; 14.2	32.3; 46.6 ; 41.3	1.5; 0.8 ; 1.3	2.6; 5.9 ; 2.0	1.0; 0.5 ; 1.6	1.3; 3.0 ; 1.1	77.2; 157 ; 33.7
	pine	58.4; 43.1 ; 73.3	7.7; 6.2; 7.2	49.2; 103 ; 54.7	116; 40.7 ; 38.8	5.0; 3.9 ; 4.4	7.7; 9.2 ; 7.0	6.7; 3.3 ; 7.2	5.9; 5.4 ; 5.7	236; 130 ; 135
	larch	156; 11.0 ; 89.7	12.8; 4.5 ; 9.4	41.1; 82.3 ; 53.9	81.6; 30.8 ; 54.0	9.3; 1.3 ; 5.5	14.6; 5.9 ; 11.0	20.6; 1.8 ; 9.6	11.9; 4.0 ; 6.4	187; 99.5 ; 132
Ząbkowice (n=8)	birch	7.5; <0.1 ; 25.0	3.3; 3.9 ; 5.3	47.1; 61.4 ; 54.7	26.1; 22.9 ; 32.9	1.0; 0.3 ; 2.0	4.8; 4.3 ; 4.5	3.3; <0.1 ; 6.1	4.0; 0.5 ; 4.6	34.1; 14.1 ; 46.3
	oak	19.1; <0.1 ; 12.3	3.9; 1.9 ; 3.6	30.1; 49.0 ; 22.3	26.2; 13.8 ; 24.0	1.7; 0.2 ; 1.2	4.2; 1.8 ; 1.9	6.3; <0.1 ; 4.1	4.8; 1.0 ; 2.8	43.7; 20.8 ; 30.5
	pine	48.0; <0.1 ; 58.0	7.0; 2.9 ; 8.4	81.3; 255 ; 111	42.1; 24.0 ; 46.6	4.5; 0.5 ; 4.4	8.7; 2.6 ; 7.5	16.3; 0.2 ; 19.5	13.5; 2.6 ; 9.3	147; 62.6 ; 74.3
	larch	108; 18.2 ; 86.1	11.9; 3.6 ; 10.9	102; 137 ; 92.8	63.9; 28.6 ; 53.5	8.3; 1.7 ; 6.4	12.7; 4.9 ; 9.6	37.8; 4.8 ; 9.6	19.4; 4.1 ; 14.2	155; 83.7 ; 136
Radunia (n=8)	birch	0.7; <0.1 ; 22.8	23.9; 16.9 ; 26.9	33.2; 24.2 ; 24.6	432; 480 ; 350	0.6; 0.6 ; 3.0	4.2; 7.3 ; 3.1	0.6; <0.1 ; 4.6	3.0; 0.9 ; 5.7	69.1; 66.8 ; 52.8
	oak	7.1; <0.1 ; 21.9	13.0; 12.8 ; 8.3	14.8; 17.9 ; 11.2	207; 429 ; 159	1.5; 0.3 ; 1.1	1.4; 1.3 ; 2.3	1.5; <0.1 ; 3.8	2.3; 1.3 ; 3.1	23.8; 25.7 ; 18.1
	pine	6.5; <0.1 ; 24.8	68.3; 56.4 ; 41.7	48.1; 95.7 ; 50.3	237; 271 ; 165	2.0; 1.2 ; 3.4	1.5; 1.0 ; 3.7	1.1; 0.4 ; 6.9	2.8; 1.1 ; 6.8	24.8; 25.8 ; 79.9
	larch	16.8; <0.1 ; 29.2	20.9; 14.5 ; 24.2	13.5; 35.1 ; 29.0	378; 183 ; 140	6.4; 0.3 ; 2.7	2.9; 0.6 ; 4.3	3.2; <0.1 ; 5.8	5.9; 0.3 ; 5.4	41.7; 31.9 ; 173
Złoty Stok (n=8)	birch	4.3; <0.1 ; 3.2	9.1; 7.5 ; 9.8	90.0; 43.8 ; 121	42.6; 25.0 ; 43.3	0.8; 0.2 ; 0.6	6.1; 5.4 ; 1.0	3.9; <0.1 ; 35.0	2.5; 0.2; 1.1	21.6; 28.3 ; 24.6
	aspen	14.4; 1.8 ; 5.9	5.1; 2.3 ; 2.8	113; 142 ; 98.8	26.8; 19.4 ; 16.7	1.5; 0.5 ; 0.6	5.6; 1.4 ; 0.7	3.3; 0.5 ; 0.8	4.1; 3.4 ; 1.8	81.6; 121 ; 16.3
	pine	14.4; <0.1 ; 15.4	7.2; 6.3 ; 5.8	29.7; 45.8 ; 13.8	30.9; 20.3 ; 30.6	1.7; 0.7 ; 1.9	3.8; 4.7 ; 4.2	4.9; 1.8 ; 4.6	3.7; 0.8; 2.4	29.8; 22.9 ; 54.3
	spruce	17.3; 0.4 ; 6.1	6.2; 3.9 ; 3.9	6.1; 11.2 ; 6.8	42.1; 24.9 ; 34.3	2.1; 0.3 ; 0.6	4.8; 1.9 ; 3.5	4.0; 0.1 ; 0.9	4.1; 3.4 ; 1.8	37.3; 53.8 ; 22.9
Przasnysz (n=8)	birch	1.4; 14.4 ; 1.5	14.2; 14.9 ; 13.7	23.2; 28.7 ; 22.9	31.3; 73.8 ; 64.2	0.3; 0.4 ; 0.2	0.6; 1.2 ; 0.7	0.2; 2.3 ; 0.3	1.2; 2.0 ; 1.3	11.7; 13.9 ; 6.9
	oak	<0.1 ; 1.6 ; 1.1	4.0; 3.7 ; 3.3	4.0; 5.0 ; 7.5	32.6; 30.0 ; 37.8	0.1; 0.1 ; 0.2	0.3; 0.4 ; 0.6	<0.1 ; 0.1 <0.1 ; 0.1	0.6; 0.8 ; 1.1	15.4; 5.9 ; 6.2
	pine	<0.1 ; 34.5 ; 8.2	5.5; 6.0 ; 9.2	38.7; 35.2 ; 23.2	30.3; 31.0 ; 35.9	0.2; 0.8 ; 0.4	0.4; 2.1 ; 0.9	<0.1 ; 6.1 ; 0.6	0.9; 3.1 ; 2.1	9.0; 18.8 ; 12.3
	larch	<0.1 ; 2.3 ; 13.5	9.2; 5.2 ; 4.7	23.7; 24.7 ; 13.2	50.5; 36.7 ; 37.3	0.6; 0.2 ; 0.6	0.9; 0.7 ; 1.3	<0.1 ; 0.4 ; 1.3	1.9; 1.1; 2.5	49.1; 11.6 ; 17.2

kg), while in bark it was fourth (27 g/kg), after K (9.3%) and Mg (5.0%). Zn was ~43 g/kg in wood and twigs and 30 g/kg in bark. Cu in wood was high (577 mg/kg) (Fig. 3, third row, Table S1). Cr reached 236 mg/kg in twigs and >100 mg/kg in bark and wood. The highest

Ni concentration among all biomass samples collected at this site was recorded in pine twigs (116 mg/kg). Relatively high Cd contents (~40 mg/kg), comparable to those in deciduous biomass, were observed in bark and wood samples (Table 4).

Fig. 3 Chemical composition charts for ash samples after combustion of wood biomass collected in Łosień. Capital letters indicate tree species (B – birch, O – oak, L – larch, P – pine), whereas lowercase letters indicate the type of biomass (t – twigs, w – wood, b – bark). The percentage is related to the total mass of 14 investigated main components



In combusted larch biomass, Ca ranged from 45.7% in twigs to 57.1% in bark. Fe was very high, reaching ~40% in twigs and 22% in bark (232 and 137 g/kg, respectively). S was also elevated (>30 g/kg in twigs and bark; 27.5 g/kg in wood) and was the third most abundant macroelement in twigs and bark, while in wood K (15%) and Mg (13.2%) dominated. In bark, K and Mg did not exceed 7% and 2.5%, respectively, and in twigs both were <2%. Excluding wood, Al was high (21 g/kg in twigs, 12 g/kg in bark), while wood showed elevated Mn (17 g/kg). Zn ranged from 52 to 56 g/kg in bark and twigs and 62–66 g/kg in wood. Cu was highest

in wood (~730 mg/kg) (Fig. 3, fourth row, Table S1). Twigs showed high Cr (182–192 mg/kg), V (154–159 mg/kg), Mo (~14 mg/kg), W (4.25–4.5 mg/kg), and Pb (~1540 mg/kg). Cd was highest in wood and bark (52–83 mg/kg) (Table 4 and S1).

3.4 Ząbkowice

In pine samples, Ca accounted for ~73% in wood and bark and 56% in twigs. Mg and K were also significant in wood, with all three elements constituting 91% of the composition.

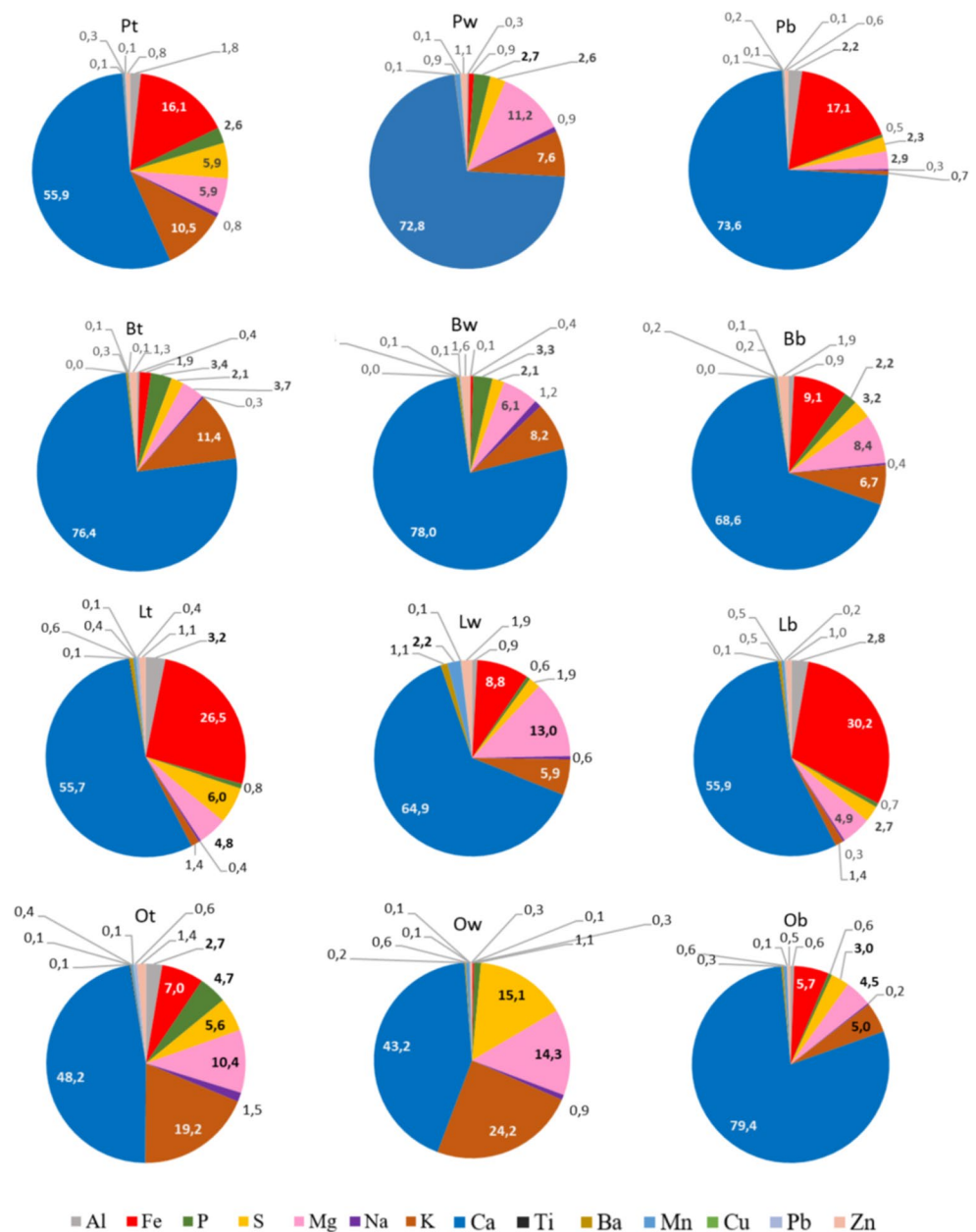
Zn in wood ranged from 5.5 to 5.8 g/kg, while Cd reached up to 260 mg/kg, similar to Nowy Dwór. In bark and twigs, Fe was the second most abundant element (95 and 92 g/kg, ~17%), and together with Ca exceeded 90% in bark. Twigs also contained K, Mg, and S (>5%). Zn was lower in twigs (4.4 g/kg) and bark (3.5 g/kg) (Fig. 4, first row, Table S1). Cu ranged from 200 to 250 mg/kg, while Cr (> 140 mg/kg) occurred only in twigs (Table 4 and S1).

In birch biomass ash, Ca ranged from 69% to 78%. In wood ash, K (>47 g/kg), Mg (36 g/kg), and P (19 g/kg) were also significant, together exceeding 95% of macronutrients. Twig samples showed higher K-to-Mg ratios (11.4% vs. 3.7%), while P remained similar to wood. Bark had elevated Fe (>56 g/kg, >9%) and higher Mg than K. All samples

contained high Zn (7.7–>11 g/kg). Wood Zn was comparable to other sites. Bark also showed high Cu (~400 mg/kg) (Fig. 4, second row, Table S1).

The Ca content in larch biomass ash was lower than in birch and pine, reaching 65% in wood and 55% in twig and bark samples. Mg content in wood was relatively high (75–77.5 g/kg), accounting for 13% of macronutrients, while Fe (8.8%) and K (5.9%) were also notable. In twig and bark samples, Fe was much higher (26.5% and 30%, respectively), corresponding to 156 and 172 g/kg. Twig samples also contained high S (>35 g/kg, 6%). Mg in twigs and bark was 4.8–4.9%. Zn content in wood was similar to birch (~2%). Wood showed elevated Mn (~13 g/kg), while twig and bark contained higher Al (19–16 g/kg) (Fig. 4, third

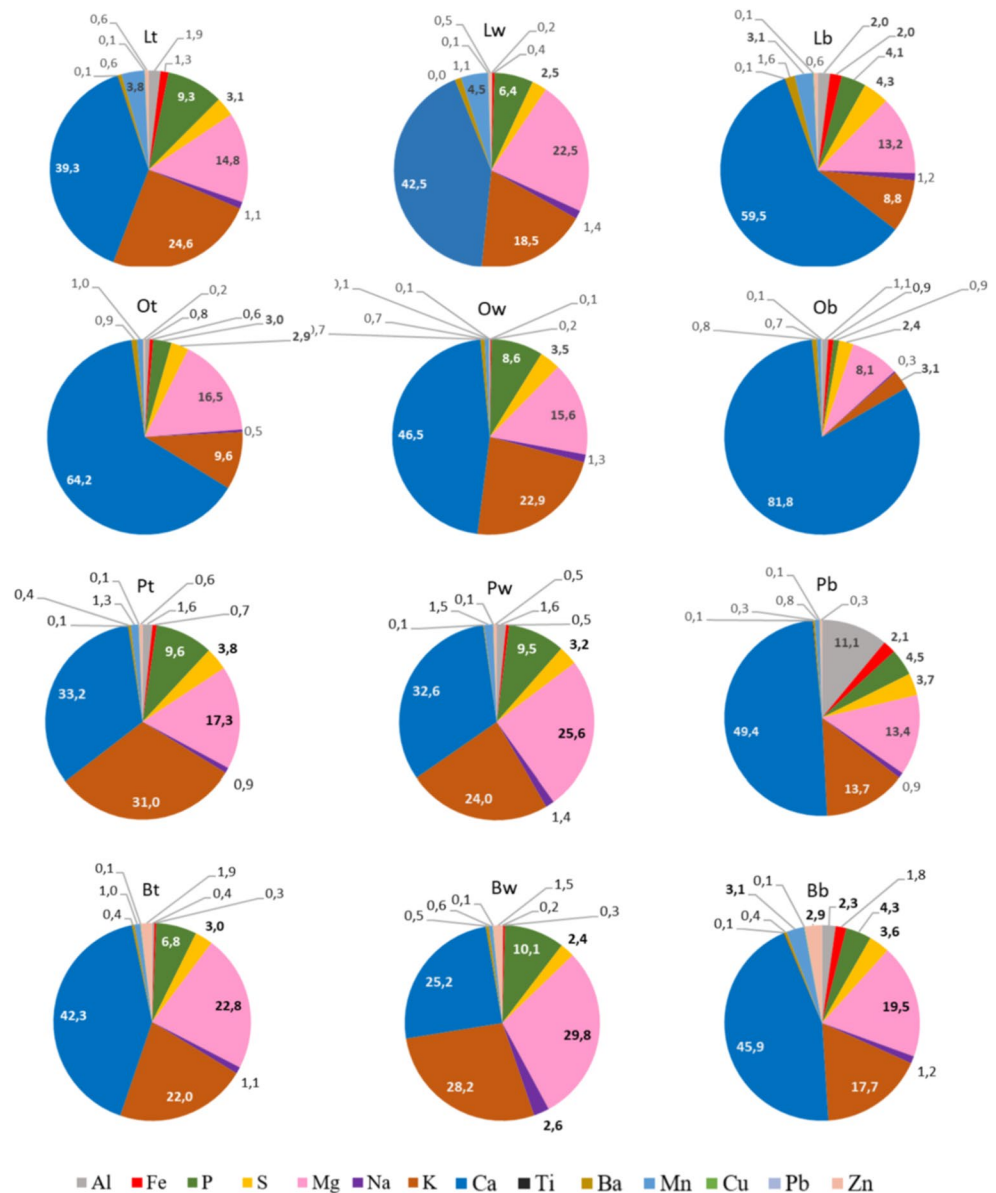
Fig. 4 Chemical composition charts for ash samples after combustion of wood biomass collected in Zabkowice. Capital letters indicate tree species (B – birch, O – oak, L – larch, P – pine), whereas lowercase letters indicate the type of biomass (t – twigs, w – wood, b – bark). The percentage is related to the total mass of 14 investigated main components



row, Table S1). Twig samples were also enriched in trace elements: Cu (~410 mg/kg), Cr (155 mg/kg), V (109 mg/kg), Cd (101 mg/kg), Mo (12.5 mg/kg), and W (8.3 mg/kg). Similar but lower levels occurred in bark. Tungsten levels were among the highest observed in all analyzed biomass (Table 4 and S1).

In oak biomass, Ca dominated only in bark (>79%), while in other fractions it was <50%. In wood, K was significant (>24%), with S (15%) and Mg (14%) also high; S reached ~96 g/kg, the highest among samples. In twigs, K was 19%, and Fe, S, and P accounted for 7%, 5.6%, and 4.7%, respectively. In bark, Fe, K, and Mg were similar (4.5–5%). Zn in wood was low (~1.8 g/kg) compared to other species, and twig Zn was similar to pine. Other PTEs were also lower than in the other species (Fig. 4, fourth row, Table S1).

Fig. 5 Chemical composition charts for ash samples after combustion of wood biomass collected in Radunia. Capital letters indicate tree species (B – birch, O – oak, L – larch, P – pine), whereas lowercase letters indicate the type of biomass (t – twigs, w – wood, b – bark). The percentage is related to the total mass of 14 investigated main components



3.5 Radunia

In larch biomass ash from the Radunia Nature Reserve, Ca was highest in bark (~60%), compared to 40% in twigs and 42% in wood. Mg and K were the next most abundant elements; Mg exceeded 22% in wood and was 14% and 13% in twigs and bark, respectively (>140 g/kg in wood). Only in twigs did K exceed Mg (~25%). Phosphorus and manganese were also elevated: P ranged from 53 to 55 g/kg (9.3% in twigs) to ~21 g/kg in bark (~4%), while Mn ranged from 3.1% in bark to 4.5% in wood (5.5–29 g/kg). Sulphur varied from 2.5% in wood to 4.3% in bark. Cu in bark reached ~600 mg/kg (Fig. 5, first row, Table S1). Ni was very high in all samples (150 mg/kg in bark to 480 mg/kg in twigs), as were Co (20–25 mg/kg) and Cr (140 mg/kg in twigs), as were Co (20–25 mg/kg) and Cr (140 mg/kg in twigs).

kg in bark to >380 mg/kg in twigs; 174 mg/kg in bark, the highest). W exceeded 6 mg/kg in twigs (Table 4).

In oak bark, Ca accounted for ~82%, while Mg exceeded 8.1%; K and S were 3.1% and 2.4%, respectively. In twigs, Ca was ~64%, with Mg (16.5%), K (9.6%), and S and P (~3% each). In wood, Ca was 46.5%, followed by K (23%) and Mg (15.5%). P was high in wood (51 g/kg; 8.6%), while S was similar across fractions (Fig. 5, second row, Table S1). Ni was very high in wood (430 mg/kg), exceeded 200 mg/kg in twigs, and reached 160 mg/kg in bark. Cu in wood exceeded 380 mg/kg (Table 4 and S1).

Pine biomass showed variable macro- and micronutrient contents. Ca did not exceed 50%, while twigs had similar Ca and K levels (165 and 154 g/kg; 33% and 31%). Wood showed similar Ca, with K and Mg at 25% each. Bark had high Al (52 g/kg; 11%). Wood and twigs had high P (9.5%), S ranged 3.2–3.8% (Fig. 5 third row, Table S1). Ni was 272 mg/kg (wood) and 236 mg/kg (twigs), lower in bark (152–175 mg/kg). Co exceeded 55 mg/kg, and Cu was >425 mg/kg in wood and 330–350 mg/kg in other samples (Table 4 and Table S1).

In birch biomass samples, Ca was lower than in other species, ranging from 46% in bark to 25% in wood. In wood, Mg dominated (~30%), followed by K (28%) and Ca (25%); Mg exceeded 165 g/kg, the highest value among all samples. P accounted for ~10%. In twigs, Ca was slightly over 42%, while Mg and K were ~22% each; P was 37 g/kg (6.8%). In bark, Mg and K decreased to 19.5% and 17.7%, and P to 4.3%. S ranged from 2.5 to 3.5% across all samples. Mn reached 140 g/kg in bark (3.7%). Zn was high in all fractions (>13 g/kg in bark, 11 g/kg in twigs, 8.7 g/kg in wood) (Fig. 5 fourth row, Table S1). Ni was very high (>450 mg/kg in wood and twigs, 350 mg/kg in bark), as was Cu (>450 mg/kg). Co was ~24 mg/kg in bark and twigs (Table 4 and S1).

3.6 Złoty Stok

The main component of birch was Ca, ranging from ~70% in twigs to 49% in wood. Birch wood contained 152–155 g/kg of K (>30% of macronutrients), while its share in twigs and bark was 9–12.5%. Mg was similar, with higher relative share in twigs (>10%) than K. Mn was highest in bark (88 g/kg; ~12%), the maximum in woody biomass; in wood and twigs it was 2.7–3.1%. Na in bark exceeded 28 g/kg (3.8%). Zn ranged from 12.5 g/kg in wood to >19 g/kg in twigs (Fig. 6, first row, Table S1). Bark also contained Cd (119–122 mg/kg) and Sb (35 mg/kg) (Table 4).

In spruce biomass from this site, Ca content exceeded 50% only in bark. However, K content was high, ranging from 23% in both bark and wood to over 30% in twigs. Mg content ranged from 7% to nearly 10% in twigs. Phosphorus content was high, particularly in wood, where it exceeded

45 g/kg (9.7%). The highest Ba content was recorded in bark samples, exceeding 18 g/kg, while in wood and twigs it exceeded 14 and 10 g/kg, respectively. Zn content in ash samples after combustion of spruce biomass ranged from 5 to 6 g/kg (Fig. 6, third row, Table S1).

The Ca share in combusted aspen twigs and bark samples was high, reaching 74% and 82%, respectively, while in wood it accounted for only 50%. K, P, and Mg contents were also high in aspen wood (136 g/kg, >75 g/kg, and 73 g/kg, respectively). In bark samples, K and Mg shares were only 12.5% and 7.5%, respectively, while in twig samples their combined share exceeded 10%. Zn content in aspen wood was also high (>12 g/kg), as were Cd (>140 mg/kg) and Cr (>120 mg/kg). Zn content in branch samples was lower but still high, reaching >8 g/kg, and in bark slightly below 6 g/kg (Fig. 6, fourth row, Table 4 and S1).

3.7 Przasnysz

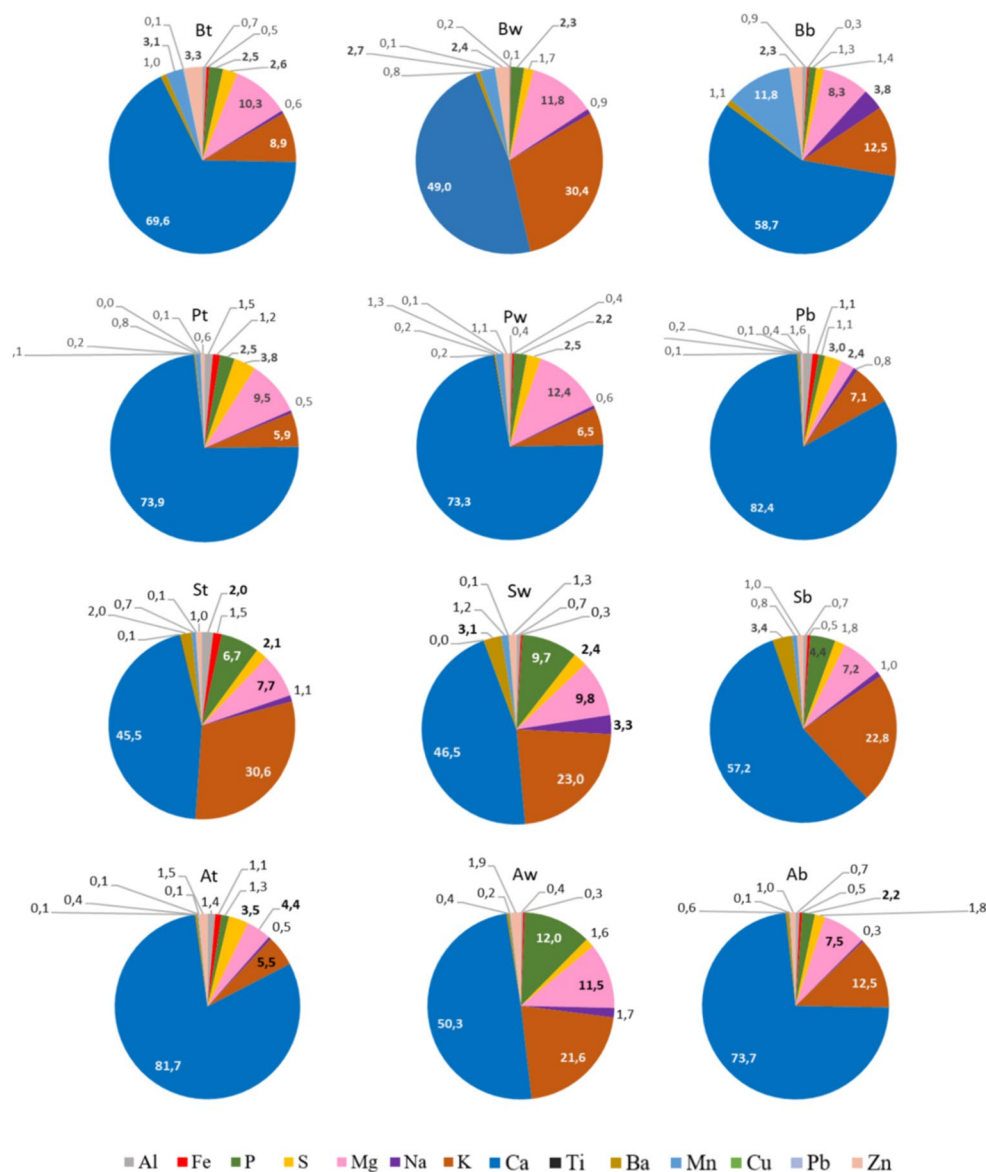
The Przasnysz site in northern Poland served as a reference area. Ca dominated ash samples after biomass combustion, ranging in birch from 55% in bark to >77% in wood. In twigs, K and Mg followed (14.6% and 6.7%), while in bark K, P, and Mg reached 18.2%, 9.5%, and 8.4%. Mn was very high, exceeding 30 g/kg in all samples and >45 g/kg in wood, accounting for 8.4% of the composition and ranking second after Ca. S ranged from 14 to 18.5 g/kg. Zn exceeded 10 g/kg in twigs and bark and 9 g/kg in wood (Fig. 7, first row, Table S1). Birch wood also showed high Cu (425–436 mg/kg) and Ni (>70 mg/kg) (Table 4 and S1).

In pine biomass samples, Ca was the dominant component, with its share ranging from slightly below 50% in bark to almost 70% in wood. K exceeded 20% in bark and accounted for approximately 10% in the remaining samples. Mg content ranged between 10% and 12%. Among the remaining components, similarly to birch, significant contributions were made by Mn (>40 g/kg in twig samples) and P (>33 g/kg in bark). Al content in wood was also relatively high (>43 g/kg), while S content ranged from 12 to 20 g/kg (Fig. 7, second row, Table S1).

The Ca content in oak wood and twigs was the highest among all biomass samples, reaching >600 g/kg (>86%) and >420 g/kg in bark. K, Mn, and Mg each accounted for more than 2%, while P and S were detected also in bark (Fig. 7, third row, table S1).

Larch samples showed different nutrient levels. Ca exceeded 60% in bark and was just over 36% in twigs. K ranged from 13% (bark) to 21% (twigs), and Mg from 7.6% to over 16%. P content was high (88–91 g/kg in twigs, ~14%; 53–55 g/kg in wood). Mn ranged from 23 to 46 g/kg (4.6–7%), S from 2% to 3%, and Cu reached up to 450 mg/kg (Fig. 7, fourth row, Table S1).

Fig. 6 Chemical composition charts for ash samples after combustion of wood biomass collected in Złoty Stok. Capital letters indicate tree species (B – birch, A – aspen, S – spruce, P – pine), whereas lowercase letters indicate the type of biomass (t – twigs, w – wood, b – bark). The percentage is related to the total mass of 14 investigated main components



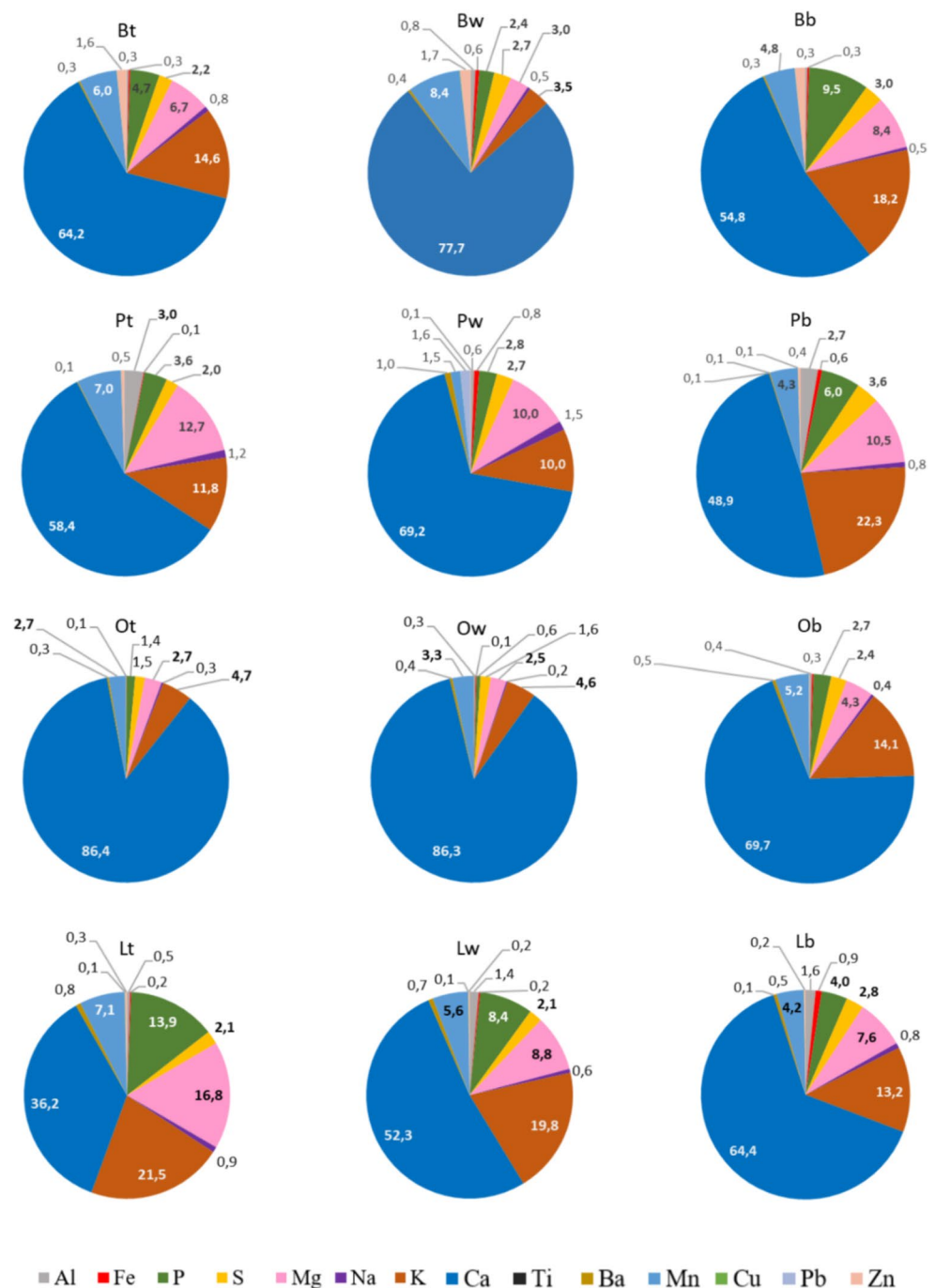
4 Discussion

Biomass sampling was performed separately, with wood samples collected from the trunk or thick branches, as well as from bark and small twigs, in order to determine the sources of elements in the biomass. It can be generally assumed that most elements found in wood originate from the soil and weathered bedrock, whereas elements present in bark and small twigs—which, in the case of coniferous species, also included needles—are primarily derived from air pollution associated with the deposition of industrial and urban dust.

Regarding trees growing on waste dumps, the substrate composition directly influences the chemical profile of the woody biomass. The flotation waste dump at the Nowy Dwór site is highly enriched with Zn and Pb ore mining

residues, containing also significant amounts of Cd, Sb, Sn, and Fe (Nadłonek et al. 2024). According to previous research (Strzyszczyński and Heinkele 2002), the Zn concentration in the waste material ranges from 19,370 to 39,500 mg/kg. Such high Zn concentrations in the substrate resulted in elevated Zn concentrations in the ash obtained from birch, larch, poplar, and pine wood. This region is also characterized by long-term industrial and urban dust emissions associated not only with zinc–lead smelting but also with other industrial activities, including hard coal combustion, iron smelting, and coke production (Rachwał et al. 2020). Suschka and Zielonka (1995) reported that Pb deposition in the Bytom area reached 2,822 mg/m²/year during the 1990s. Although the total atmospheric deposition of metals and metalloids in the Bytom area has decreased substantially to 40–64 mg/m²/year (Michalski and Pecyna-Utylska 2022),

Fig. 7 Chemical composition charts for ash samples after combustion of wood biomass collected in Przasnysz. Capital letters indicate tree species (B – birch, O – oak, L – larch, P – pine), whereas lowercase letters indicate the type of biomass (t – twigs, w – wood, b – bark). The percentage is related to the total mass of 14 investigated main components



secondary dust emissions from degraded land may still constitute a significant source of air pollution.

At Złoty Stok, the waste originates from arsenic ore mining and flotation and contains, in addition to As (which was not analyzed in the present study), significant amounts of Fe, S, and Pb (Karczewska et al. 2007). The elevated Pb concentrations observed in the ash of twigs and bark of coniferous trees at the Nowy Dwór site, indicate atmospheric deposition, including redeposition resulting from dusting from sparsely vegetated surfaces, is the main source

of this element in the area. Larch twigs, needles, and bark collected at the Ząbkowice site, as well as at Łosień, located 4.5 km away, also accumulated high Pb concentrations, are most likely associated with dust deposition from a nearby ore-bearing dolomite quarry. Larch twigs and bark, and to a lesser extent pine, at the Nowy Dwór, Ząbkowice, and Łosień sites also accumulated other trace elements, including Sb and Sn. In contrast, significantly lower Pb concentrations were found in larch wood confirming that outer tissues are more exposed to direct deposition. The accumulation

of PTEs deposited on conifer needles has previously been described by Steinnes (1989), and conifer needles have been widely used as biomonitors of atmospheric pollution in several countries (Steinnes 1989; Čeburnis and Steinnes 2000). A different pattern was observed for pine and birch, for which significantly higher Pb concentrations were measured in wood than in the other biomass fractions at the Nowy Dwór waste dump, suggesting effective root uptake under the specific conditions prevailing in the technogenic soils.

Cadmium exhibited a different distribution pattern, with the highest concentrations observed in pine and poplar wood at the Nowy Dwór site and in pine wood at Łosień. In pine twigs and bark, Cd concentrations were generally lower, with the exception of poplar bark at the Nowy Dwór site, where atmospheric deposition, including dust resuspension from sparsely vegetated surfaces, appears to play a significant role. Despite the high concentrations of both Pb and Cd in the flotation tailings, the high soil pH limits Pb bioavailability. In contrast, Cd, which is considerably more mobile in the soil environment, is more readily taken up by plants and accumulates to a greater extent in woody biomass, particularly in species with softer wood, such as pine and poplar. The ability of poplar wood to accumulate Cd has previously been reported for contaminated soils in Central Europe (Unterbrunner et al. 2007). Softwood tree species, particularly fast-growing species such as willow, poplar, birch, and pine, generally exhibit a greater capacity to take up and accumulate heavy metals from contaminated soils than many hardwood species (Lovynska et al. 2025).

In all tree species growing in Upper Silesia, except oak, high Zn concentrations were recorded in wood, bark, and twigs. Notably, extremely high Zn concentrations were observed in birch wood at the Nowy Dwór site and at the other study sites, regardless of soil Zn concentrations or atmospheric deposition rates. Birch has previously been identified as a Zn hyperaccumulator, which is reflected in elevated Zn concentrations in its wood, leaves, and bark (Çomaklı and Bingöl 2021; Fidos and Rutkowska 2023; Jurković et al. 2023). Similar patterns were observed in all birch biomass samples collected across the studied sites. At the second waste dump, Złoty Stok, developed on arsenic ore mining and flotation waste, high Zn concentrations were also found in all aspen biomass fractions, particularly in wood, as well as in pine wood. The concentrations of other trace elements at Złoty Stok remained moderate, with the exception of Cr in aspen wood, where concentrations exceeded 100 mg/kg and were comparable to those measured in other tree species growing on the Nowy Dwór waste dump. The highest Cu concentrations were restricted to the wood of coniferous species growing at Łosień and Nowy Dwór, pointing to localized industrial sources.

Evidence that natural substrates can significantly influence the accumulation of PTEs in woody biomass is provided by the results obtained at the Ząbkowice and Radunia sites. The latter, located within a nature reserve on the summit of the Ślęza Massif, is situated in a region with relatively low atmospheric pollution and is therefore exposed primarily to the long-range transport of pollutants. However, the soil, developed on a metal-rich serpentinite substrate, is likely responsible for the observed accumulation patterns (Koner et al. 2023).

Elevated Ni concentrations, recorded only in larch and pine twigs and needles at Łosień and were attributed to local atmospheric deposition from the iron and steel industry. At the Radunia site (with the exception of larch), the highest Ni concentrations were found in wood samples, suggesting direct uptake from the substrate. Biomass fractions (wood, bark, and twigs) also contained significantly higher Co and W concentrations than those collected at the other sites. Mg concentrations in wood were likewise very high, which is consistent with the abundance of Mg in the minerals composing serpentinite. In contrast, at Ząbkowice, where the parent material is dolomite, Mg concentrations in wood were considerably lower. Only the oak wood sample exhibited Mg concentrations comparable to those recorded at the Radunia site. Very high S concentrations were also observed in oak wood from Ząbkowice. The availability of elements to trees at these two sites is likely related to the marked difference in soil pH between Ząbkowice (alkaline soil, pH 8.0) and Radunia (acidic soil, pH 4.8), which is one of the main factors controlling the availability of nutrients and metals to plants. The low soil pH, extending into deeper soil horizons at the Radunia Reserve, together with the high geogenic Ni and Cr concentrations, has been confirmed by previous studies (Szuszkiewicz et al. 2016; Kierczak et al. 2021). A similar pattern has been reported for soils developed on serpentinite in Asia, where Mg concentrations are significantly higher than those of other major cations (Yang et al. 2022). This is related to the abundance of Mg-rich minerals in the parent material, such as olivine, serpentine, and chlorite, which are readily susceptible to weathering.

Two of the studied sites were located on natural Podzols developed on Quaternary sands. These sites differed markedly in their levels of contamination. Łosień, situated in an industrialized area in the immediate vicinity of a metallurgical plant and a coke plant, was characterized by high Fe concentrations accompanied by elevated concentrations of Zn, V, Ni, Mo, Cr, Cd, and W. An important observation is that Fe and most PTEs (with the exception of Zn, Cd, and Cu) were predominantly accumulated in twigs and bark, particularly in coniferous species (larch and pine). This distribution pattern strongly suggests an atmospheric origin. However,

long-term industrial activity has also likely contributed to the accumulation of these elements in the soil (Schaaf et al. 2004; Zhou et al., 2019). This appears to be particularly true for Zn, Cd, and Cu, whose elevated concentrations in wood suggest that long-term accumulation in the soil has enhanced their uptake by trees, especially birch.

The Przasnysz site, located in northern Poland, is situated far from local emission sources in a region characterized by relatively low historical atmospheric deposition (Table 3). It was therefore considered the reference site for the other study locations. Nevertheless, very high Mn concentrations were recorded in all biomass fractions at Przasnysz and, similarly to the other sites, very high Zn concentrations were observed in birch biomass. Given the absence of major industrial and urban emission sources, together with the low concentrations of these metals in the upper 30 cm of soil (Table 3), the most plausible explanation is the presence of an *ortstein* layer in deeper soil horizons, a feature commonly observed in Podzols of northern Poland (Jankowski 2014). This interpretation is supported by the characteristic elemental composition of *ortstein* horizons described in the literature (Lundström et al. 2000), which are often enriched in elements released during mineral weathering and subsequently immobilized during podzolization. Under acidic and humid conditions, Mn and Al are more mobile than Fe and may therefore be taken up by plant roots (Lipiec et al. 2018). High concentrations of these elements were detected in the wood of the studied tree species, whereas only very low concentrations of typically anthropogenic trace elements were found in wood, bark, and twigs. Particularly noteworthy were the very high P concentrations in larch wood and especially in needles. Larch accumulates P primarily in actively growing tissues and storage organs, with the highest concentrations occurring in young tissues such as needles and current-year shoots (Zeng et al. 2018). Similarly, Mn accumulates mainly in metabolically active tissues, including young shoots and twigs, but also in older tissues such as wood. Mn is an essential plant micronutrient and a constituent of numerous enzymes; however, its availability increases under acidic soil conditions, promoting root uptake. Excess Mn may subsequently be detoxified through sequestration in tissues that are periodically shed, including needles (Wildová et al. 2021). Nevertheless, confirmation that the *ortstein* layer is the primary source of these metals in tree biomass requires further investigation, including deep soil excavations.

Species-level analysis of woody biomass revealed substantial variation in the concentrations of both nutrients and PTEs among the studied tree species and among different biomass fractions. Birch should be excluded

as a source of biomass ash intended for soil fertilization because, regardless of location, soil, bedrock, or air pollution, it exhibited hyperaccumulation of zinc and, to a lesser extent, manganese and copper. Silver birch is highly tolerant of elevated Zn concentrations. In polluted areas and on waste dumps, it can accumulate more than 4,000 mg/kg of Zn in its leaves. Although such concentrations rarely pose a direct threat to the tree itself, they may pose a significant risk to the surrounding ecosystem and human health (Balafrej et al. 2020; Krupnova et al. 2021). Coniferous species growing in polluted areas accumulated substantial amounts of pollutants in twigs, needles, and bark, although some species also accumulated considerable amounts of PTEs in wood. These species may therefore be useful for biomonitoring atmospheric pollution. Larch appears to be particularly suitable for this purpose because it sheds its needles annually and may therefore serve as an indicator of year-to-year changes in air pollution during the growing season (May–October). Larch accumulated the highest concentrations of pollutants in its needles; however, at some sites (Radunia and Przasnysz), elevated concentrations of certain elements may also have originated from the soil parent material. The lowest concentrations of pollutants were observed in oak wood, with the exception of Ni and Cr, whose concentrations were elevated at sites where the parent material or deposited dust was enriched in these elements. Compared with a similar study conducted in the Czech Republic (Asare and Hejzman 2022), substantially higher PTE concentrations were recorded in most tree species examined in Poland. However, the Czech study included only two sites, both of which were less contaminated than those investigated in the present study. Nevertheless, ash obtained from the combustion of birch, larch, pine, and oak wood exceeded permissible limits for Pb or Zn at both sites.

In Poland, the permissible concentrations of selected PTEs in fertilizers are regulated by the Act of 10 July 2007 on Fertilizers and Fertilization. Most of the analyzed biomass ash samples exceeded the legal limits for Cd, Pb, Cr, or Ni specified in this legislation. Only biomass combustion ash derived from Przasnysz (excluding birch) met the requirements for use as a fertilizer in agricultural or forest soils.

Moreover, the combustion of contaminated wood in fireplaces or domestic heating systems not equipped with electrostatic precipitators may pose an environmental risk, as volatile potentially toxic metals, such as Zn, Cu, Pb, and Cd, become concentrated in the finest fraction of particulate matter emitted with the flue gases through chimneys (Magiera et al. 2025).

5 Conclusions

The elemental composition of tree biomass is highly dependent on the specific tree species and the tissue type. Tree wood primarily serves as an indicator of soil and bedrock composition, whereas bark, twigs, and needles predominantly accumulate atmospheric pollutants from industrial and urban dust deposition.

Metal accumulation in forest biomass is driven by a combination of atmospheric deposition, historical industrial activities, and natural geogenic backgrounds. Soil pH acts as a primary regulatory mechanism, where acidic soils enhance the mobilization and root uptake of geogenic metals like nickel and manganese, while alkaline soils limit their bioavailability.

Coniferous species, particularly larch, serve as highly effective biomonitors for seasonal air quality due to the annual shedding and high pollutant retention capacity of their needles.

Silver birch functions as a potent hyperaccumulator of zinc across all environmental conditions. This innate physiological trait results in an extreme concentration of zinc in its tissues, rendering biomass ash derived from birch systematically unsuitable for environmental recycling or soil fertilization.

Heavy metals display distinct allocation patterns within woody biomass. Highly mobile elements like cadmium are readily absorbed via roots and transferred into internal wood tissues, whereas less mobile elements like lead remain heavily concentrated in outer, atmosphere-exposed tissues such as bark and needles.

Biomass harvested from post-industrial, technogenic, or metal-rich geological areas yields combustion ash that consistently violates statutory heavy metal thresholds. Except for biomass sourced from clean reference areas, wood ash from these regions cannot be safely used as a fertilizer and must be managed as regulated waste to prevent domestic particulate pollution and environmental re-contamination.

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Author contributions T Magiera wrote the main manuscript text and prepared fall tables and interpretation of geological, geochemical and soil data. A Łukasik prepared all figures and a part of discussion related to the interpretations of forest data and species differences in the accumulation of macro- and microelements.

Data availability The data were deposited in RepOD data repository and will be available after the embargo expired at 31. January 2028 at the following URL: <https://doi.org/10.18150/OP00P7>.

Declarations

Competing interests The authors declare no competing interests.

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